# MAGNETIC TORQUE*
# Mτ1-A
## STUDENT LAB MANUAL

A PRODUCT OF TEACHSPIN, INC.

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# ***Introduction to the Instructor***

This student lab manual for the *M$\tau$-1A* Magnetic Torque instrument serves a dual purpose. The first is to direct students to discover experimentally the physical laws that govern a magnetic dipole's interaction with a magnetic field. The second is to direct students toward developing the laboratory technique required to accomplish this. The key verb in both of these goals is "direct". Through asking the student specific questions rather than dictating specific instructions, we have sought to lead the student along the correct experimental pathway, while at the same time allowing choices that will make the student contemplate and understand the physics involved. We believe that this will result in greater retention of theoretical and experimental physical principles in the minds of the students.

We envision this lab manual being used in the following manner. The week before the first laboratory session that utilizes the Magnetic Torque instrument, the lab instructor will take three to five minutes to familiarize the students with the instrument, its accessories, and its capabilities. The students will then receive the chosen *Pre-Lab Assignment*, which they will complete before the next laboratory session. This *Pre-Lab Assignment* thoroughly covers all aspects of the theory involved in the experiment. By completing the *Pre-Lab Assignment* and understanding all of the questions and answers included in it, the students will be able to come into the lab and perform the experiment. In order to check that the students have completed and understand the *Pre-Lab Assignment*, the lab instructor may wish to give a short quiz at the beginning of the lab session. Knowing that a quiz will be given helps motivate the students to prepare.

While in the laboratory, the students should have with them their *Lab Help-Sheet* and *Pre-Lab Assignment*. The *Lab Help-Sheet* gives helpful hints and questions that direct the students toward the use of good experimental technique, while allowing for choices to be made.

A lab write-up may or may not be necessary. If data is recorded and plotted, and calculations made, during the laboratory period, then the tables, graphs, and a short explanation of the result may be all that is required for the lab instructor to evaluate the students' progress. However, more extensive write-ups are certainly possible and do give the students the opportunity to develop scientific writing skills.

We hope that this lab manual will help you in your implementation of the experiments involved with the Magnetic Torque instrument, and will also help to teach your students about the laws and investigation of the physical universe.

# ***Static Torque Experiment*** ---- *Pre-Lab Assignment*

OBJECTIVE:

A magnetic dipole experiences a net torque in a uniform magnetic field. Take advantage of this physical principle to determine the magnetic moment of the dipole which is inside the cue ball given to you by your instructor.

QUESTIONS:

1. The picture below shows the Magnetic Torque cue ball floating on the air bearing in an applied uniform magnetic field, ***B***. This corresponds to the condition when the currents in the top and bottom coils of the Magnetic Torque instrument are flowing in the same direction. In what direction are the currents flowing if the magnetic field is pointing upward?

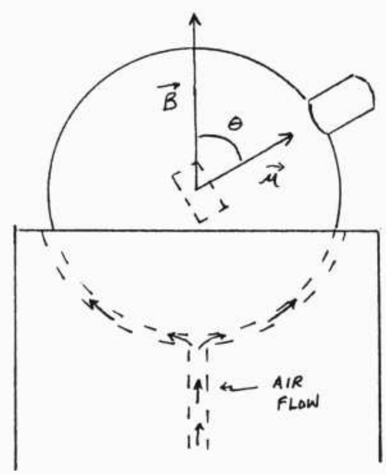



Figure 1

2. The magnetic moment, $\mu$, is shown as a vector, as is the magnetic field. Draw on Figure 1 any magnetic torque experienced by the ball, and give the magnitude of this torque in algebraic form.

3. The picture on the next page again shows the ball floating on the air bearing in an applied uniform magnetic field. This time, however, an aluminum rod of mass *M* carrying a moveable plastic cylindrical weight of mass *m* has been inserted into the ball. Identify all of the torques experienced by the ball, and give the magnitude of those torques in algebraic form. Do all of the torques act in the same direction? How do we add or subtract torques? (When you're in the lab, you will observe that in the absence of a magnetic field, and without the extra rod, the cue ball is nearly rotationally balanced. Neither the handle nor the embedded magnet produce a significant gravitational torque.)

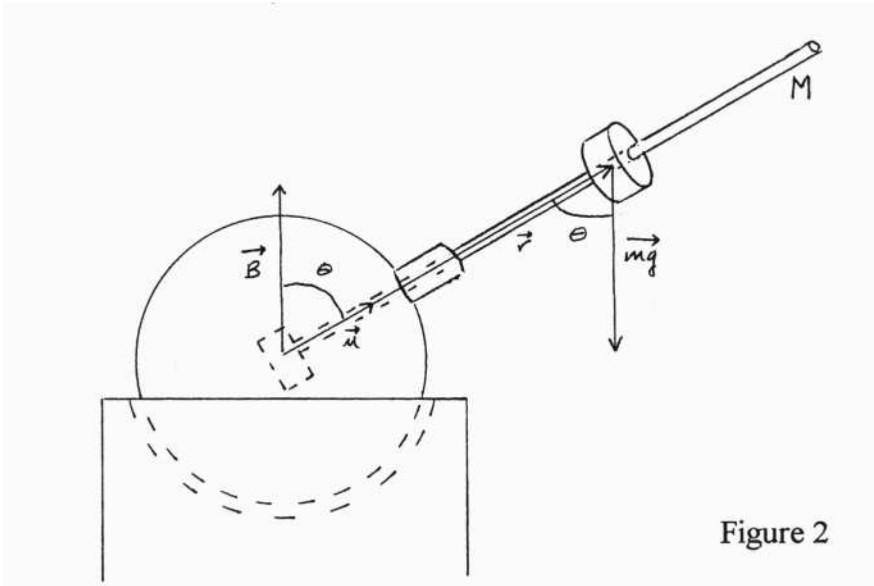



Figure 2

4. If the magnitude of the magnetic torque is greater than the gravitational torque in the situation shown in Figure 2, will there be a net torque on the ball?  If so, will this torque cause the ball to begin to rotate in a particular direction?

5. If the magnitude of the magnetic torque is less than that of the gravitational torque, will there be a net torque on the ball?  If so, will this torque cause the ball to begin to rotate in a particular direction?

6. If the magnitude of the magnetic torque is equal to the magnitude of the gravitational torque, what will be the net torque experienced by the ball?  Will the ball rotate?  Write down this physical situation in algebraic form.

7. In the equation in Question #5, what letters represent known quantities, or quantities that we can directly measure using the Magnetic Torque instrument and standard laboratory equipment?  (Hint:  The magnitude of the magnetic field is directly proportional to the current, with the proportionality factor being 0.00137 teslas/ampere.)

8. In the equation in Question #5, what is the remaining symbol representing the unknown constant quantity that is not directly measurable?

9. There are two possible methods by which we could calculate this unknown constant quantity. The first is by making a “single-point” measurement. While the ball is in static equilibrium, we could make one measurement of each of the directly measurable quantities involved in our equation, and from those single measurements calculate the unknown constant quantity. But notice that there are two directly measurable quantities that the experimenter can vary. What are these two variables?

10. Since there are two variables, we can make multiple measurements, and from these multiple measurements calculate the unknown constant quantity. This is what we will call a “multiple-point” measurement. If we were to change one of the experimenter-controlled variables, how would we have to adjust the magnitude of the second variable in order to maintain rotational static equilibrium?

11. If we plotted the first variable with respect to the second variable, what kind of graph would we expect to obtain? Sketch a hypothetical example of such a graph and label the axes.

12. How could we extract the value of the unknown constant quantity from this graph?

13. Which method would yield a more reliable and accurate value for the unknown constant: the "single-point" or "multiple-point" measurement? Explain your reasoning.

14. Take a moment to review the previous eleven questions. You should be able to begin to develop an experimental strategy for accomplishing the stated objective. To solidify this strategy, construct a table, which you can use in the laboratory to record the data needed. Be sure to construct your table so that data can easily be transferred from tabular to graphical form.

# ***Static Torque* Experiment** ---- *Lab Help-Sheet*

The following is a list of helpful hints and questions that you should look over prior to your lab session and take into the laboratory with you. They will help you to think about and identify good experimental techniques.

REMEMBER: The final objective of the experiment is to determine the unknown magnetic moment of your given dipole.

1. Useful tools:
   - calipers calibrated to at least 1 mm
   - ruler calibrated to at least 1 mm
   - scale calibrated to at least one-tenth of a gram

2. Bring into the laboratory the data tables, which you have constructed, but feel free to modify them if you discover a better way of organizing your data.

3. In addition to recording your data in tabular form, it is important to plot your data on graph paper as you perform the experiment. This ensures that if an aberrant data point appears, you will be able to recognize it and repeat the measurement to investigate the validity of that point.

4. Does it matter whether the magnetic field is the dependent or independent variable in the experiment?

5. The mass of the rod and location of its center of mass are constant throughout the experiment. Will these values affect the slope of the straight line data plot? If not, what characteristic of the plot will they affect?

6. Does performing one trial of the experiment or multiple trials of the experiment lead to greater confidence in your result?

7. Does having more data points on the graph lead to greater confidence in your result?

8. REMINDER: The magnetic field is directly proportional to the current as read on the Magnetic Torque ammeter, with the proportionality factor being 0.00137 teslas/amp. To verify this value, approximate the coil magnet as two co-axial current loops, each having a radius of 9.9 cm, and separated by 14.0 cm. Keep in mind that the coils have 195 turns each, and the Magnetic Torque ammeter measures the current passing through one turn. Since coils are in series, the current passing through each turn is identical. However, what if the coils were in parallel? Would the current necessarily be identical in each coil?

SET UP A TABLE TO RECORD YOUR DATA. MAKE GRAPHS USING GRAPH PAPER. SHOW SAMPLE CALCULATIONS. EXPLAIN YOUR RESULT.

# ***Harmonic Oscillation Experiment*** ---- *Pre-Lab Assignment*

OBJECTIVE:

A magnetic dipole in a uniform magnetic field experiences a net magnetic torque. This torque can act as a restoring torque on a physical pendulum. Take advantage of this physical principle to determine the unknown value of the magnetic moment of a given dipole, when it is part of a harmonic oscillating physical pendulum.

QUESTIONS:

1. Shown below is a picture of the Magnetic Torque cue ball (with its embedded magnetic dipole) floating on the air bearing in a uniform magnetic field. This corresponds to the condition when the currents in the top and bottom coils of the Magnetic Torque instrument are flowing in the same direction. Draw on the diagram any torques experienced by the ball, and give the magnitude of those torques in algebraic form. (When you're in the lab, you will observe that in the absence of a magnetic field the cue ball is nearly rotationally balanced. Neither the handle nor the embedded magnet contribute any significant gravitational torque.)

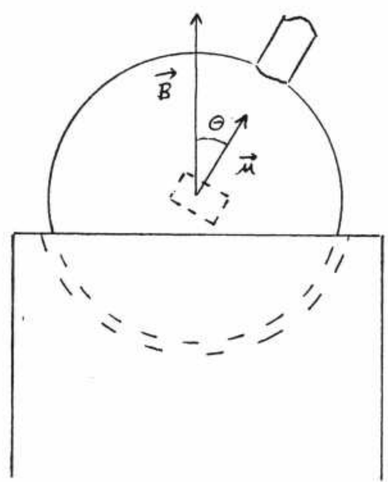



Figure 1

2. How is a net torque on a body related to that body's change in angular momentum? Write this relationship in algebraic form.

3. Given your answer to question #2, will the instantaneous angular acceleration of the handle of the ball in Figure 1 be in the clockwise or counterclockwise direction?

4. Suppose the ball in Figure 1 was just released from rest. In which direction will the ball begin to rotate?

5. Now suppose that the ball is rotating in a counterclockwise direction around an axis perpendicular to the page. As shown in the picture below, the magnetic moment has an instantaneous angular displacement to the left of the vertical. Draw in the torque experienced by the ball at this instant, and give the magnitude of this torque in algebraic form.

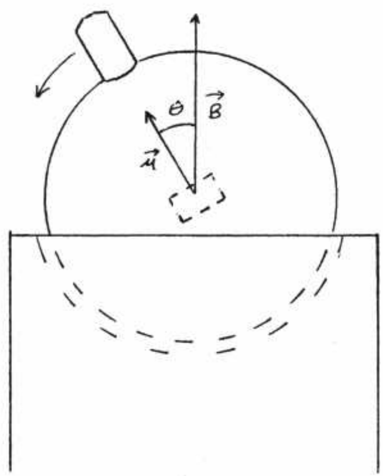



Figure 2

6. Given the relationship between torque and the change in angular momentum established in question #2, is the instantaneous angular acceleration of the ball in Figure 2 in the clockwise or counterclockwise direction?

7. Does the torque in Figure 2 support or oppose an increase in the angular displacement of the handle?

8. For our physical situation, $\tau = -\mu B \sin\theta$ . The negative sign is present since the torque is a restoring torque. Also, $L = I\frac{d\theta}{dt}$. $I$ is the moment of inertia of the ball, which we can approximate as that of a sphere. Thus $I = \frac{2}{5}mr^2$. Insert these expressions for torque and angular momentum into the differential equation obtained in question #2, leaving the moment of inertia as $I$.

9. For small angle displacement, $\sin\theta \approx \theta$. Make this substitution in the equation from question #8.

10. Look carefully at the equation that you just wrote down. Have you seen any other equation that has a form similar to this one? (Hint: a mass on a spring.) What kind of motion does this differential equation describe?

11. Look at the equation from question #9. For a given small angle, what happens to the magnitude of the angular acceleration as the magnetic field increases in magnitude?

12. The expression $\theta = A\sin\,\omega t$ is the solution for the differential equation you wrote in question #9. In this solution, $A$ and $\omega$ are unknown constants. Substitute this expression into the differential equation and solve for $\omega$, the angular frequency of oscillation. What is the physical significance of $A$?

13. Write an algebraic statement of the relationship between $\omega$, the angular frequency, and $T$, the period of oscillation. Use this to rewrite the equation from question #12 in terms of $T$ rather than $\omega$, and solve for $T^2$.

14. Look at your result from question #13. What letters in this equation represent physical quantities that can be measured using the Magnetic Torque instrument and/or standard laboratory equipment? (Hint: The magnetic field is directly proportional to the current with the proportionality factor being 0.00137 teslas/ampere.)

15. Of the letters identified in question #14, some represent quantities that can be varied by the experimenter, while others represent quantities that are constant throughout the whole experiment. What letters fall into each category?

16. What quantity does the remaining symbol, the unknown constant, represent?

17. Suppose we varied one of the variables from question #15, and then recorded the values of the second variable as we varied the first variable. Is there some way to plot our data in order to extract the value of the magnetic moment from the graph? (Hint: Straight line graphs are often helpful.)

18. Take a moment to review the previous seventeen questions. After doing so, you should be able to begin to develop an experimental strategy for obtaining the stated objective. Solidify this strategy by preparing a table that will be used in the laboratory to record the needed data. Make sure that the table is constructed so that your data can easily be transferred from tabular to graphical form.

# ***Harmonic Oscillation Experiment*** ---- *Lab Help-Sheet*

The following is a list of helpful hints and questions that you should look over prior to your lab session and take into the laboratory with you. They will help you to think about and identify good experimental techniques.

REMEMBER: The final objective of this experiment is to determine the unknown magnetic moment of your given dipole.

1. Bring your data tables into the laboratory, but feel free to modify them if you discover a better way of organizing your data.

2. In addition to recording your data in tabular form, it is important to plot your data on graph paper as you perform the experiment. This ensures that if an aberrant data point appears you will be able to recognize it and repeat the measurement to investigate the validity of that point.

3. Useful tools:
   - stopwatch
   - calipers calibrated to 1 mm
   - scale calibrated to one tenth of a gram

4. Is it easier to set $B$ as the independent variable, or $T$ ?

5. Which produces more confidence and accuracy in your period measurement: timing one period or multiple periods? How many periods seem reasonable? Is there a practical limit?

6. Which produces more evenly spaced data points on your straight-line graph: 1.) making more period measurements at higher magnetic fields than at lower fields; 2.) making more period measurements at lower fields than at higher fields; or 3.) making the same amount of measurements at low, medium, and high fields?

7. REMINDER: The magnetic field is related to the applied current as read on the Magnetic Torque ammeter by the value 0.00137 teslas/amp.

SET UP TABLES TO RECORD DATA. MAKE GRAPHS USING GRAPH PAPER. SHOW SAMPLE CALCULATIONS. EXPLAIN YOUR RESULT.

# ***Precessional Motion Experiment*** ----*Pre-Lab Assignment*

OBJECTIVE:

A magnetic dipole experiences a net torque in a uniform magnetic field. Under certain conditions, this torque will cause a rotating body to precess. Take advantage of this physical principle to determine the unknown magnetic moment of a given dipole.

QUESTIONS:

1. The picture below shows the Magnetic Torque cue ball floating on the air bearing in a uniform magnetic field. This corresponds to the condition when the currents in the top and bottom coils of the Magnetic Torque instrument are flowing in the same direction. Draw on the picture any torques that the ball experiences, and give the magnitude of these torques in algebraic form. (When you're in the lab, you will observe that in the absence of a magnetic field the cue ball is nearly rotationally balanced. Neither the handle nor the embedded magnet produce a significant gravitational torque.)

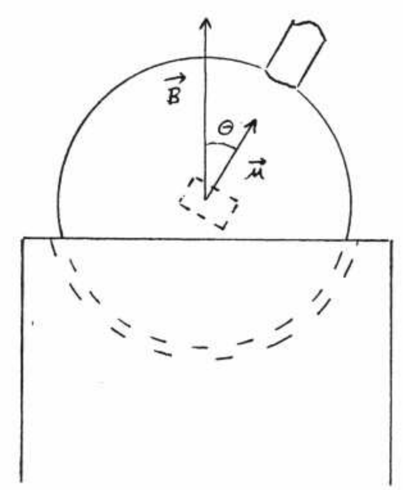



Figure 1

2. What relationship exists between a torque on a body and that body's change in angular momentum? Express this relationship in algebraic form.

3. Suppose that the ball in Figure 1 was just released from rest. Based on your answer to question #2, describe the motion of the ball.

4. There should be some mention of angular momentum in your answer to question #3. Is the angular momentum vector parallel, anti-parallel, or perpendicular to the torque vector in the situation shown in Figure 2?

5. Does the motion described in question #3 involve changes in the magnitude of the angular momentum? Does the motion involve changes in the direction of the angular momentum?

6. Now suppose that the ball is given a large spin angular momentum about its handle's axis before being released from rest, as shown in the picture below. Is the torque experienced by the ball parallel, anti-parallel, or perpendicular to the spin angular momentum?

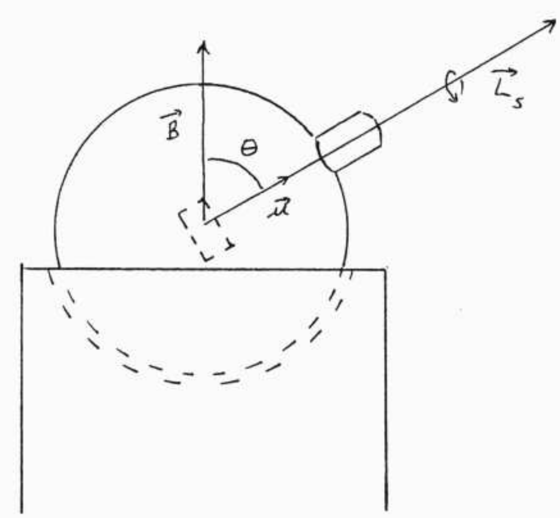



Figure 2

7. Will the torque cause a change in the magnitude of the spin angular momentum? Will it cause a change in the direction of the spin angular momentum?

8. Describe the motion of the ball in Figure 2. (Hint: think of how this situation is similar to a gyroscope in the earth's gravitational field.)

9. Carefully read through the supplemental derivation of the expression for the precessional period of the ball. Look carefully at the final expression. What letters represent physical quantities which we can measure using standard laboratory equipment and/or the Magnetic Torque instrument? (Hint: The magnitude of the magnetic field is directly proportional to the current, with the proportionality factor being 0.00137 teslas/ampere.)

10. What letters represent quantities that can be *varied* and measured? What letters represent quantities that the experimenter can keep constant, at least during the time it takes to make each separate measurement? Note that you, as the experimenter, may have some options in this experiment as to which quantities to vary and which to keep constant. Make sure to justify your decision.

11. What is the remaining letter that represents an unknown constant quantity?

12. What kind of graph (somehow incorporating the two variables on the two axes) would we like to obtain in order to extract the value of this unknown constant quantity?

13. How would you then extract this unknown constant quantity from the graph?

14. Take a moment to review the previous thirteen questions. After doing so, you should be able to begin to develop an experimental strategy that can be used to achieve the stated objective. Solidify this strategy by constructing a table that can be used in the laboratory to record your needed data. Make sure that the table allows you to easily transfer data from tabular to graphical form.

# ***Precessional Motion Experiment*** ---- *Lab Help-Sheet*

The following are some helpful hints and questions that you should read over before the lab session and take into the laboratory with you. They will help you to think about and identify good laboratory techniques.

REMEMBER: The experiment's final objective is to determine the unknown magnetic moment of your given dipole.

1. Bring into the lab the data tables, which you have constructed, but feel free to modify them if you encounter a better way of organizing your data.

2. In addition to recording your data in tabular form, it is important to plot your data on graph paper as you perform the experiment. This ensures that if an aberrant data point appears, you will be able to recognize it and repeat the measurement to investigate the validity of that point.

3. Useful tools:
   - calipers
   - scale
   - stopwatch
   - strobe light set to "ON" on Magnetic Torque front panel

4. How do you know when the ball's spin frequency is equal to the strobe light frequency?

5. Is it easier to vary the precessional period ($T$) or the magnetic field ($B$) as the independent variable?

6. Can you devise a way to keep the ball's spin frequency constant for all measurements throughout the experiment? Is this necessary in your particular experimental strategy?

7. Which yields more confidence in your result and is experimentally easiest to obtain: a low, medium, or high constant spin frequency of the ball?

8. Should there be current passing through the magnet coils continuously throughout the experiment, or should you just turn the current on and off as you need it?

9. Should you time the ball's precession for one period, or for multiple periods?

10. Should more data be collected for low or high magnetic fields, or equal amounts for both?

11. Does the direction of the field seem to matter for obtaining a better result?

12. Is the angular frequency of the ball's spin equal to the frequency as read off the strobe display?

13. REMINDER: The magnetic field is related to the current as read on the Magnetic Torque ammeter by the value 0.00137 teslas/amp.

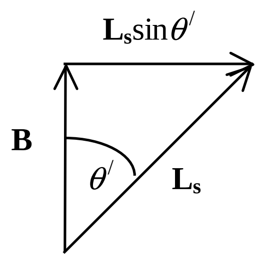
SET UP TABLES TO RECORD DATA. MAKE GRAPHS USING GRAPH PAPER. SHOW SAMPLE CALCULATIONS. EXPLAIN YOUR RESULT.

<u>*Supplement to Precessional Motion Experiment*</u> ---- *Derivation of Precessional Period*

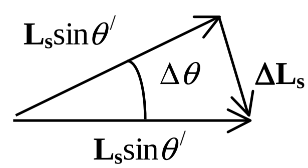
The differential equation for the rotational motion of the ball is:

$$\boldsymbol{\tau} = \frac{d\mathbf{L}}{dt}$$

A side view of the angular momentum vectors is shown below:




If we look from above the spinning ball at the change of the angular momentum for a short time $\Delta t$, the picture looks like the one below:




Since $s = r\theta$, we can show that:

$$\Delta L_s = \Delta\theta\, L_s \sin\theta'$$

$$\frac{\Delta L}{\Delta t} = \frac{\Delta\theta}{\Delta t} L \sin\theta'$$

as $\Delta t \rightarrow 0$,

$$\frac{dL}{dt} = \Omega_p L \sin\theta'$$

where $\Omega_p$ is the precessional angular velocity. From our original differential equation,

$$\frac{dL}{dt} = \mu B \sin\theta'$$

So we obtain the expression

$$\mu B \sin\theta' = \Omega_p L \sin\theta'$$

$$\mu B = \Omega_p L$$

$$\Omega_p = \frac{\mu}{L} B$$

Since

$$\Omega_p = \frac{2\pi}{T_p}$$

and

$$L = I\omega$$

we obtain the expression

$$\boxed{T_p = \frac{2\pi I\omega}{\mu}\frac{1}{B}}$$

# ***Field Gradient Experiment*** ---- *Pre-Lab Assignment*

OBJECTIVE:

A magnetic dipole experiences a net force in the presence of a magnetic field gradient. Take advantage of this physical principle to determine the unknown magnetic moment of a given dipole.

QUESTIONS:

1. Consider the picture below that shows a magnetic dipole suspended from a spring in a uniform magnetic field. This corresponds to the condition when the currents in the top and bottom coils of the Magnetic Torque instrument are flowing in the same direction. The dipole is free to rotate in its housing. Does the dipole experience a net torque? Does it experience a net force? Draw in any torques or forces as vectors. (Hint: remember that a dipole can be modeled as a current loop.)

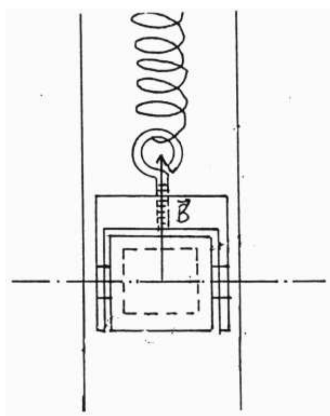



Figure 1

2. Consider now the following picture of a dipole suspended from a spring in a *non*-uniform magnetic field, with the magnitude of the magnetic field at the dipole is zero. This corresponds to the condition when the currents in the top and bottom coils of the Magnetic Torque instrument are flowing in *opposite directions*. Does the dipole experience a net torque? Does it experience a net force?

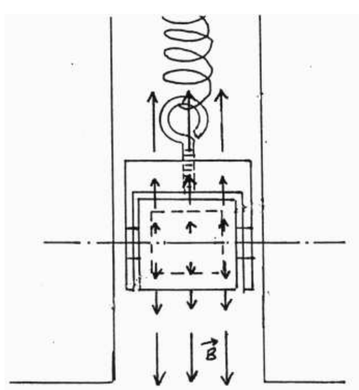



Figure 2

3. If you answered question #2 correctly, you know that in a non-uniform magnetic field, a magnetic dipole experiences a net force. If we label the vertical axis of our drawing the *z*-axis, the algebraic form of this physical principle is $F_z = \mu \frac{dB}{dz}$ . Carefully look at this equation and draw in the force vector in Figure 2.

4. We can experimentally vary the field gradient, since it is directly proportional to the current passing through the magnet coils, with the proportionality constant being 0.0193 teslas/ampere-meter. The magnetic moment is the constant that we wish to determine. Can we then measure the force? (Hint: the dipole is attached to a spring.)

5. What is the expression for the force on a spring with respect to its displacement from its equilibrium position? Substitute this expression for the force in the equation given in question #3.

6. Look at the expression obtained from question #5. There are two symbols that represent quantities which will vary and can be measured by the experimenter using the Magnetic Torque instrument and standard laboratory equipment. What are these two symbols?

7. Look again at the expression obtained from question #5. There are two symbols that remain. Since the value of the magnetic moment is our final objective in the experiment, we need to be able to measure the spring constant. How could we accomplish this measurement? (Hint: If you don't know, continue on to the next question.)

8. Consider the picture on the following page that shows a magnetic dipole suspended from a spring. Our goal is to determine the spring constant. If a small weight is added to the system, will the spring's length change?

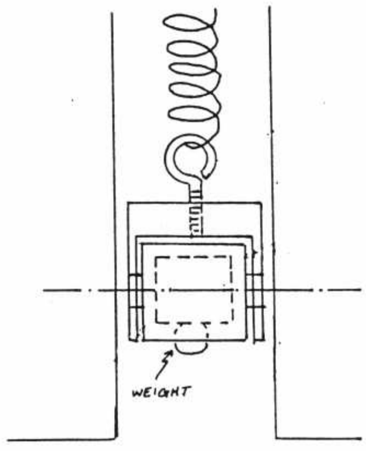



Figure 3

9. By how much will a spring's length change for an additional mass suspended from the spring? Give an expression in algebraic form that describes your answer.

10. We have a number of weights that we can add on to the magnetic dipole in varying amounts. In light of this, look at the expression from question #9. What letter represents a known physical constant in that equation?

11. Which letters represent quantities that will vary and can be measured using standard laboratory equipment?

12. What unknown constant remains? Can you make a graph involving the two variables from which you could obtain the value of this constant?

13. Take a moment to review the previous twelve questions. After doing so, you should be able to begin to develop an experimental strategy with which you can measure the sprint constant of your given spring. Solidify this strategy by constructing a data table that will be used to record the needed data.

14. Let us return back to the expression obtained in question #5. To review, what letters represent quantities that will vary and can be measured? What letter represents a quantity that is constant throughout the experiment, but can be measured?

15. What is the remaining letter which represents an unknown constant quantity?

16. Can you plot a set of data points involving the variables which would enable you to extract this unknown quantity from the graph?

17. Take a moment to review the previous fifteen questions. After doing so, you should be able to begin to develop an experimental strategy with which to obtain the stated objective. Solidify this strategy by constructing a data table that will be used to record your needed data. Make sure to construct the tables so that data can easily be transferred from tabular to graphical form.

# ***Field Gradient Experiment*** ---- *Lab Help-Sheet*

Look over the following hints and questions before your lab session. Be to take them into the laboratory with you. They will help you identify good experimental techniques.

REMEMBER: The final objective of the experiment is to determine the unknown magnetic moment of your given dipole.

1. Bring your data tables into the laboratory with you, but feel free to modify them if you encounter a better way to organize your data.

2. In addition to recording your data in tabular form, it is important to plot your data on graph paper as you perform the experiment. This ensures that if an aberrant data point appears, you will be able to recognize it and repeat the measurement to investigate the validity of that point.

3. Useful tools: ruler, balance

4. What marking on the tower's calibrated scale indicates the center of the coil magnet?

5. Is the magnitude of the field gradient necessarily the same at the center of the coils as it is some distance above or below the center?

6. Does the field direction (whether it is up or down) affect your result? Does one setting allow for easier measurements?

7. Which is easier: to use the field gradient (which depends on the current) or the spring displacement as the independent variable?

8. When the dipole is displaced downward, is it easier to make displacement measurements using the scale on the tower, or rather to return the dipole to its zero position and then use a ruler to measure the change in length of the rod that projects from above the tower?

9. In light of your answer to question #3, when the dipole is displaced upward, is it better to make displacement measurements using the tower scale, or rather to return the dipole to its zero position and then use a ruler to measure the change in length of the rod that projects from above the tower?

10. REMINDER: The field gradient is proportional to the current as read on the Magnetic Torque ammeter, with the proportionality constant being 0.0193 teslas/amp-meter. To verify this value, approximate the coil magnet as two co-axial current loops of opposite direction (one clockwise, one counterclockwise), each having a radius of 9.9 cm, and separated by a distance of 14.0 cm. Keep in mind that the coils have 195 turns each, and the current read on the ammeter is the current through one of those turns.

SET UP TABLES TO RECORD DATA. MAKE GRAPHS USING GRAPH PAPER. SHOW SAMPLE CALCULATIONS. EXPLAIN YOUR RESULT.